

OxyGuard[®]

Model 420

4-20 mA Two-Wire

Dissolved Oxygen Probe

OWNER'S MANUAL

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1. DESCRIPTION

The **Model 420 Oxygen Probe** can measure both dissolved oxygen and oxygen in gas, and can be used in fish farms, sewage treatment works and similar as well as to measure air, pure oxygen, other gases, oils, wine etc.

The **Model 420** is a genuine "passive" 2-wire measurement transmitter, i.e. when you connect it to a power supply its current consumption changes between 4 and 20 mA corresponding to the oxygen concentration it senses.

It can be calibrated so that the 4-20 mA corresponds to a wide variety of measuring ranges by placing it in air or other calibration gas and adjusting the calibration trimmer until the current is correct.

The **Model 420** also contains **galvanic isolation** circuitry, so that an oxygen measurement made with a Model 420 cannot be disturbed electrically by other measurements.

This manual covers three types of Model 420:

- Type D033M for mg/l dissolved oxygen,
- Type D033SV for % saturation dissolved oxygen or % volume oxygen in air or other gas.

Type D033V, which is used for measuring the purity of oxygen from oxygen generators.

The following uses and measuring ranges are the most common:

Waste water treatment etc.:

D033M; 4-20 mA = 0-10 mg/l.

Aquaculture, the environment and similar:

D033M; 4-20 mA = 0-20 mg/l or 0-40 mg/l.

D033SV; 4-20 mA = 0-100%, 0-200% or 0-400% saturation.

For measuring the air and oxygen in gas:

D033SV; 4-20 mA = 0-25% volume oxygen.

For measuring pure oxygen:

D033V; 4-20 mA = 0-100% volume oxygen.

The **Model 420** consists of an upper part with cable gland, electronic circuitry and calibration adjustment screw; a part with cathode and anode, and a cap with membrane and electrolyte. When measuring dissolved oxygen it functions correctly with liquid movement down to 1 cm/s.

The **Model 420** is virtually maintenance free - just wipe the membrane and check the calibration from time to time, the frequency depending on the actual conditions. **OxyGuard oxygen probes do not need regular service.**

Changing the membrane is only necessary if the membrane should be damaged or if, after long use (years) it is not possible to calibrate up to the calibration value. The procedure, which renovates the probe completely, is simple and can be performed by anyone who can follow the instructions in this manual. Spares are shipped with the probe.

The **Model 420** can be connected to a power supply and any measuring or metering equipment (PLC's, controllers, dataloggers etc) that can sense 4-20 mA as long as there is always enough voltage across the Model 420.

NB. The Model 420 can also be fitted with a temperature sensor with 4-20 mA 2-wire transmitter. The standard range is -5 to +45°C.

Model 420 can also be obtained fitted with a Pt100 or Pt1000 sensor without transmitter.

NB This manual does NOT cover probes that have a red collar on the cable just above the probe.

The tough membrane can be wiped clean with a cloth or soft paper.

2. INSTALLATION

The **OxyGuard Model 420** is easy to install. It can be hung in the lead, but OxyGuard make a range of fixing devices that afford a more secure mount, both for use in liquids or when the Model 420 is used to measure gas. Make sure that the probe is not likely to suffer mechanical disturbance or damage.

When used in liquid mount it where there is suitable movement. Approx. 1 cm/sec is sufficient at 7 mg/l and a temperature of 13°C. Do not locate it where it can move and strike against the wall of the tank or other fixture, do not locate it directly over diffusers or similar.

Electrical connection is very simple - connect the **Model 420** in series with a power supply of between 12 and 32 volts DC and any secondary instruments desired. Use the junction box provided with the unit. Make sure that the power supply output voltage is large enough to ensure that the Model 420 always has sufficient voltage across it when the loop current is a maximum. The loop current can perhaps exceed 20 mA if you calibrate by adjusting the secondary equipment (e.g. a computer or PLC span factor). See the figure below. Remember to take the resistance in cables and connections into account, and make sure that the power supply can provide power to everything connected to it. **To ensure correct operation the voltage across the Model 420 must never drop below 11 V. Any ripple in the supply must not bring the voltage below 11V.**

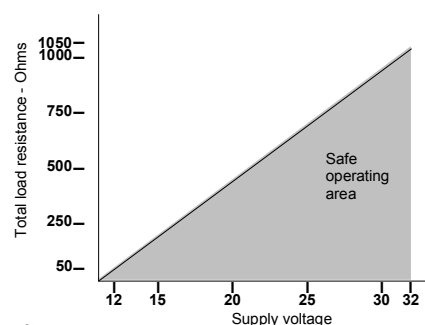
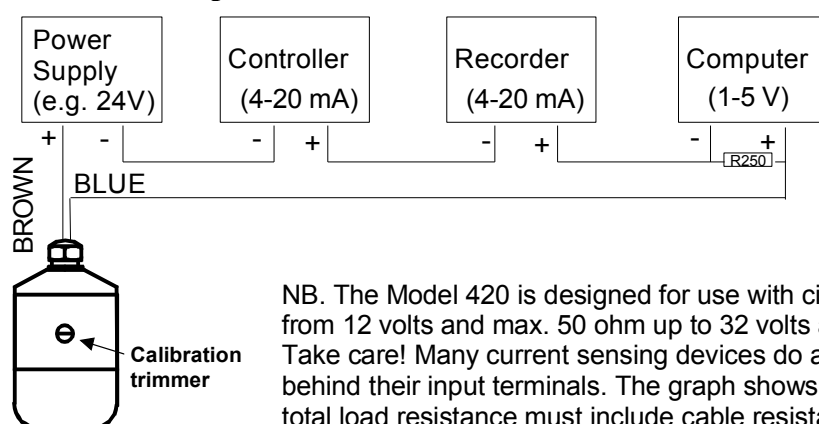
After mounting and connection the **Model 420** must be calibrated to the desired range, so that the maximum loop current of 20 mA represents the desired maximum (full scale) measurement value. The calibration procedure is described in "Directions For Use", where considerations regarding the choice of measurement range are also pointed out.

NB. The Model 420 acts as a capacitor during warm-up so secondary instruments must be capable of withstanding the resulting short current surge.

The power supply must also be capable of connection to a capacitive load; this is especially important if many Model 420's are connected to the same power supply.

NB. If you connect the Model 420 to a PLC you will probably need a 24 V DC supply. Consult OxyGuard if you are in doubt.

Connection Example



NB. The Model 420 is designed for use with circuits from 12 volts and max. 50 ohm up to 32 volts and max. 1050 ohm. Take care! Many current sensing devices do actually have resistors hidden behind their input terminals. The graph shows the safe operating area. The total load resistance must include cable resistance etc. WE RECOMMEND 24 V DC.

Lead colours:

Probes that only measure oxygen:
Brown = + oxygen
Blue = - oxygen

Probes with 4-20 mA signals for both oxygen and temperature:
Brown = + oxygen
Blue = - oxygen
Yellow = + temperature
Black = - temperature

Probes with 4-20 mA for oxygen and Pt100 or Pt1000 for temperature:
Brown = + oxygen
Blue = - oxygen
Yellow + Black = temperature 1
Beige = temperature 2

23. DIRECTIONS FOR USE

3.1 Calibration

There will usually be two possible ways of calibrating a system incorporating a Model 420: 1) by adjusting the Model 420 itself or 2) by adjusting the equipment connected to the Model 420.

When calibrating for the first time, however, you **must** choose the measurement range for the system and perform basic calibration of both the Model 420 and the equipment connected to it.

Do NOT choose too small a range, we strongly recommend that a "nice round figure" is chosen where normal measurements will lie somewhere between 8 and 16 mA. This gives excellent accuracy and still leaves scope for measurements to swing both below and above normal values. Numerical values of 0-5, 0-10, 0-25, 0-50 and 0-100 are good choices.

After having chosen the range for the system check that the equipment to be connected to the Model 420 is correctly calibrated - i.e. ensure that loop currents of both 4 and 20 mA produce the correct display or read-out.

After this you should calibrate the Model 420 itself, by placing it in liquid or gas with known oxygen content and adjusting the small screw in the side of the Model 420 until the current in the loop is correct, as in the following.

Subsequent calibration can be performed by adjusting either the Model 420 or the equipment connected.

Calibration Procedure

Calibration entails that the probe is placed in a gas or liquid with known oxygen content, and its output is then adjusted to correspond. The air is usually used, air saturated water is also used.

The probe must also have the same temperature as the air (or water) it is calibrated in, which means that the it must be allowed to adjust to the air (or water) temperature before the calibration adjustment is made. Any rugged, long-life oxygen probe like the **OxyGuard** can take up to an hour to equalize a 10°C temperature change in air, but only approx 10 minutes in water.

Before calibrating make sure that the membrane of the **Model 420** is clean. To calibrate in air wipe the membrane dry and hang the Model 420 in free air, away from direct sunlight (or wrap aluminium foil around it). When the output is stable adjust the calibration adjustment screw (trimmer) in the side of the **Model 420** until the milliamp output current has the correct value.

This value must, of course, correspond to the chosen measurement range. Whether or not air pressure, temperature and salinity affect the calibration value depends on whether you measure DO or oxygen in gas and what units you measure in, as explained in the following.

Use the same calibration values for both air and air-saturated water. Calibration against test kits is generally **not** recommended.

When measuring dissolved oxygen the **OxyGuard EasyCal** calibrator can be used to give a very fast calibration. The EasyCal establishes air-saturated conditions at the probe membrane with the probe in the water - you do not have to wait for temperature equalization in the air. The EasyCal is especially useful when calibration is performed by adjusting the read-out device or computer connected to the Model 420.

NB. Use a JTX junction box to obtain easy access to the 4-20 mA loop. Use a JTX to connect the Model 420 when you install it. Then, when you calibrate, just open the JTX and plug in a mA meter. The loop current flows through the meter instead of the JTX.

Calibration Value for Systems Measuring Dissolved Oxygen in mg/l (ppm)

The amount of oxygen that can be dissolved in water is affected by temperature, pressure and salinity. The correct calibration value for mg/l measurements must therefore be found from tables. The tables in this section show output values for ranges of 0-10, 0-20 and 0-40 mg/l, and show how you can easily calculate values for other ranges. A range of 0-5 mg/l cannot be air calibrated, and correct calibration at calibration values over 20 mA is dependant on use well within loop voltage and resistance limits so that there is always at least 12 VDC across the Model 420. It should also be noted that some secondary equipment cannot accept more than about 22 mA and will return "transmitter error" above this value.

It is essential to correct for temperature - use the air temperature near the probe. For a simple calibration at low altitudes use the table values. At high altitudes, or for greater accuracy, the calibration value should be corrected for air (barometric) pressure. The salinity should always be corrected for in salt-water measurements.

Calibration Value for Systems Measuring Dissolved Oxygen in % Saturation

If you measure % saturation and calibrate in air the calibration value is always the current that corresponds to 100% saturation. Temperature, pressure and salinity do not affect this calibration value.

It is NOT possible to air calibrate to ranges below 100% sat, this would entail adjusting to a value over 20 mA, which cannot be done.

Range	Output value
0-100%	20 mA
0-200%	12 mA
0-400%	8 mA

For other ranges calculate the output value as follows:

$$\text{Output value} = 100 \times 16 / (\text{range in } \%) + 4 \text{ mA}$$

Calibration Value for Systems Measuring % by Volume

If you measure volume % in gas at atmospheric pressure calibrate in air to the output current that corresponds to 20.9% oxygen. Note that if you measure low oxygen concentrations you can use a range of 20% and still calibrate in air. For lower ranges a calibration gas must be used.

Range	Output value
0-25%	17.376 mA

NB. % measurements, both % saturation and % volume, are affected by barometric pressure, but fluctuations in pressure seldom affect measuring accuracy more than +/- 2% of measured value.

If you measure pure oxygen (range 0-100%) then the air corresponds to a current of 7.344 mA. A special calibration gas with high oxygen content can also be used.

NB. The EasyCal for mg/l measurements has a display that shows the correct mg/l calibration value.

Temperature and pressure will not affect the accuracy of a mg/l measurement.

By definition air has the same oxygen partial pressure as water or liquid that is 100% saturated with air.

A version of the EasyCal is found without display to enable fast calibration of % saturation measurements.

IMPORTANT: Note that probes for % sat measurements have different membranes to probes for mg/l (ppm) systems.

3.3 CALIBRATION TABLE - mg/l DISSOLVED OXYGEN MEASUREMENTS

Temp °C	°F	Cal. value mg/l	Output, mA	Output, mA	Output, mA
			Range 0-10 mg/l	Range 0-20 mg/l	Range 0-40 mg/l
0	32,0	14,60	27,4	15,7	9,8
1	33,8	14,20	26,7	15,4	9,7
2	35,6	13,81	26,1	15,1	9,5
3	37,4	13,45	25,5	14,8	9,4
4	39,2	13,09	25,0	14,5	9,2
5	41,0	12,76	24,4	14,2	9,1
6	42,8	12,44	23,9	14,0	9,0
7	44,6	12,13	23,4	13,7	8,9
8	46,4	11,83	22,9	13,5	8,7
9	48,2	11,55	22,5	13,2	8,6
10	50,0	11,28	22,0	13,0	8,5
11	51,8	11,02	21,6	12,8	8,4
12	53,6	10,77	21,2	12,6	8,3
13	55,4	10,53	20,8	12,4	8,2
14	57,2	10,29	20,5	12,2	8,1
15	59,0	10,07	20,1	12,1	8,0
16	60,8	9,86	19,8	11,9	7,9
17	62,6	9,65	19,4	11,7	7,9
18	64,4	9,45	19,1	11,6	7,8
19	66,2	9,26	18,8	11,4	7,7
20	68,0	9,08	18,5	11,3	7,6
21	69,8	8,90	18,2	11,1	7,6
22	71,6	8,73	18,0	11,0	7,5
23	73,4	8,56	17,7	10,9	7,4
24	75,2	8,40	17,4	10,7	7,4
25	77,0	8,24	17,2	10,6	7,3
26	78,8	8,09	17,0	10,5	7,2
27	80,6	7,95	16,7	10,4	7,2
28	82,4	7,81	16,5	10,3	7,1
29	84,2	7,67	16,3	10,1	7,1
30	86,0	7,54	16,1	10,0	7,0
31	87,8	7,41	15,9	9,9	7,0
32	89,6	7,29	15,7	9,8	6,9
33	91,4	7,17	15,5	9,7	6,9
34	93,2	7,05	15,3	9,6	6,8
35	95,0	6,94	15,1	9,6	6,8
36	96,8	6,82	14,9	9,5	6,7
37	98,6	6,72	14,8	9,4	6,7
38	100,4	6,61	14,6	9,3	6,6
39	102,2	6,51	14,4	9,2	6,6
40	104,0	6,41	14,3	9,1	6,6

NB. The calibration values are rounded to the nearest decimal for practical reasons. This table is based on an air pressure of 760 mm Hg. Examples on the table overleaf show how to correct for salinity and other air pressure, and how to find the output value for other ranges. A range of 0-5 mg/l cannot be air calibrated, and correct calibration at calibration values over 20 mA is dependant on use well within loop voltage and resistance limits so that there is still at least 12 VDC across the Model 420. It should also be noted that some secondary equipment cannot accept more than about 22 mA and will return "transmitter error" above this value.

3.7 CALIBRATION TABLE - mg/l DISSOLVED OXYGEN IN SALT WATER APPLICATIONS

		Salinity - parts per thousand				
Temp		0	10	20	30	40
°C	°F					
0	32.0	14.6	13.6	12.7	11.9	11.1
1	33.8	14.2	13.3	12.4	11.6	10.8
2	35.6	13.8	12.9	12.1	11.3	10.6
3	37.4	13.4	12.6	11.8	11.0	10.3
4	39.2	13.1	12.3	11.5	10.7	10.0
5	41.0	12.8	11.9	11.2	10.5	9.8
6	42.8	12.4	11.6	10.9	10.2	9.6
7	44.6	12.1	11.4	10.7	10.0	9.4
8	46.4	11.8	11.1	10.4	9.8	9.4
9	48.2	11.5	10.8	10.2	9.5	8.9
10	50.0	11.3	10.6	9.9	9.3	8.7
11	51.8	11.0	10.3	9.7	9.1	8.6
12	53.6	10.8	10.1	9.5	8.9	8.4
13	55.4	10.5	9.9	9.3	8.7	8.2
14	57.2	10.3	9.7	9.1	8.6	8.0
15	59.0	10.1	9.5	8.9	8.4	7.9
16	60.8	9.9	9.3	8.7	8.2	7.7
17	62.6	9.7	9.1	8.6	8.1	7.6
18	64.4	9.5	8.9	8.4	7.9	7.4
19	66.2	9.3	8.7	8.2	7.7	7.3
20	68.0	9.1	8.6	8.1	7.6	7.2
21	69.8	8.9	8.4	7.9	7.5	7.0
22	71.6	8.7	8.2	7.8	7.3	6.9
23	73.4	8.6	8.1	7.6	7.2	7.8
24	75.2	8.4	7.9	7.5	7.1	6.7
25	77.0	8.2	7.8	7.4	7.0	6.6
26	78.8	8.1	7.6	7.2	6.8	6.5
27	80.6	7.9	7.5	7.1	6.7	6.4
28	82.4	7.8	7.4	7.0	6.6	6.2
29	84.2	7.7	7.3	6.9	6.5	6.1
30	86.0	7.5	7.1	6.8	6.4	6.1
31	87.8	7.4	7.0	6.6	6.3	6.0
32	89.6	7.3	6.9	6.5	6.2	5.9
33	91.4	7.2	6.8	6.4	6.1	5.8
34	93.2	7.0	6.7	6.3	6.0	5.7
35	95.0	6.9	6.6	6.2	5.9	5.6
36	96.8	6.8	6.5	6.1	5.8	5.5
37	98.6	6.7	6.4	6.1	5.7	5.5
38	100.4	6.6	6.3	6.0	5.7	5.4
39	102.2	6.5	6.2	5.9	5.6	5.3
40	104.0	6.4	6.1	5.8	5.5	5.2

BAROMETRIC PRESSURE CORRECTION

The table is referred to a barometric pressure of 760 mm Hg.

At other air pressures, the values should be corrected in the following way:

$$\text{Corrected value} = \text{Table value} \times \frac{\text{Actual pressure}}{760}$$

Example:

Temperature = 14°C

Salinity = 30 ppt

Air pressure = 742 mm Hg

$$\text{Corrected value} = 8.6 \times \frac{742}{760} = \underline{\underline{8.4}}$$

OTHER RANGES & CALCULATION OF OUTPUT CURRENT

To calibrate to any desired range:

- 1) Find the correct calibration value from the table,
- 2) correct it for barometric pressure if needed,
- 3) calculate the correct output current
- 4) adjust accordingly.

$$\text{Output current (mA)} = (\text{corrected cal. value} \times 16/\text{range}) + 4 \text{ mA}$$

E.G. if you calibrate at 14°C, 742 mm Hg, for water of 30 ppt salinity the correct cal. value is 8.4 mg/l as shown above.

If you have chosen a range of 20 mg/l the correct current to calibrate to will be:

$$8.4 \times 16/20 + 4 = 10.72 \text{ mA}$$

How often should calibration be performed?

Unfortunately, it is not possible to answer this question. Under ideal conditions (in air) the probe can keep its calibration for many months. In use in water the actual conditions (e.g. the nature of deposit build-up) and the desired accuracy will dictate calibration frequency. It is very important that calibration is performed with care - it is much better to perform **one** thorough calibration than two quick ones. Give the probe good time to stabilize, check the barometer and decide if correction is necessary. Check the salinity if you measure in salt water.

Try to do a good job. Remember that no measurement is more accurate than the calibration.

3.2 Maintenance

The probe's membrane must be kept free from deposits. Especially when measuring dissolved oxygen it should be cleaned at regular intervals, the frequency depending on the actual conditions. Cleaning can be performed by wiping it with a cloth or soft paper. The membrane is strong and not easily damaged, but **do not** try to scratch it clean with a fingernail! **The probe should not be taken apart unless the membrane is damaged or unless, after long use (some years), you cannot calibrate up to the correct value.** There is no need to exchange the electrolyte regularly.

4. SERVICE PROCEDURE

4.1. Probe membrane replacement and probe renovation

The probe's membrane should be wiped clean from time to time. **The probe should not be taken apart unless the membrane is damaged or unless, after long use (some years), you cannot calibrate up to the correct value.**

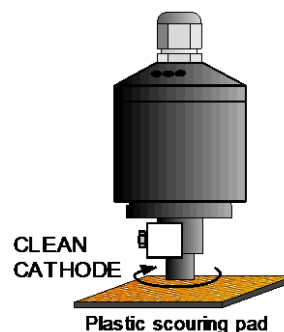
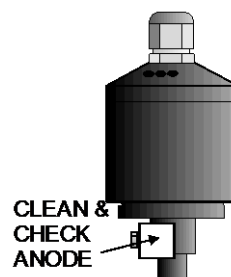
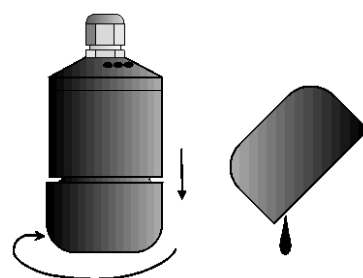
To replace the membrane and/or renovate the probe proceed as follows:

4.1.1) Remove the probe, rinse it and unscrew the cap. If it sticks, tap the side of the probe gently with a hammer then try again. Discard the electrolyte, rinse the cap and top part, clean off any brown or black oxide deposits.

4.1.2) Inspect the anode. If the probe was filled correctly when it was last renovated it will be easy to clean the dark deposits from the anode - use a nail brush or similar. If the probe was not filled completely the anode will be very corroded and must be replaced. Check that the nut under the anode is tight before fitting a new anode. Wash the anode in soapy water before use to remove any protective oil. Make sure that you use type 3 anodes on the probe.

4.1.3) Check the cathode and remove any deposits using the plastic abrasive pad supplied with the probe or a little wet or dry emery paper, grade 600. The cathode **MUST NOT BE POLISHED**.

NB: The probe can be fitted with membranes either for mg/l or for % saturation measurement. When renovating make sure you use the right membrane.



4.1.4) Rinse and dry the top part.

4.1.5) You can at this stage perform an easy check on the probe. Dry the probe – especially the cathode and area around it - completely, then connect the probe and observe the output signal – this should be 4 mA (zero). Contact your distributor if this is not the case.

4.1.6) Fill a new (or renovated) cap to the brim with electrolyte – the excess electrolyte helps remove any air bubbles.

4.1.7) Locate the flat machined from the thread. Lower the upper part into the cap and turn the cap half a turn to engage the thread.

4.1.8) Tilt the probe 15° so that the flat is uppermost and screw the cap onto the top part. Excess electrolyte and air should dribble out at the flat.

IT IS IMPORTANT THAT THE PROBE IS FILLED COMPLETELY.

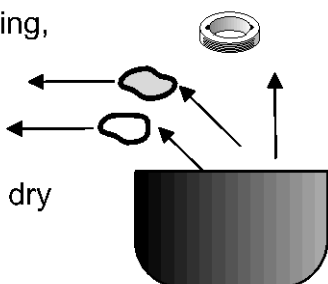
WHEN YOU ARE CERTAIN THAT THE PROBE IS FILLED COMPLETELY TIGHTEN THE CAP **HARD**.

After renovation the probe can be regarded as new, and it must be calibrated by adjusting the small trimmer in the side of the probe (see section 3 of this manual). It should be hung up in air to stabilize for at least an hour before calibration, and should be re-calibrated after a day or two, again by adjusting the calibration trimmer in the side of the probe.

A new membrane can easily be fitted to the cap - see the drawing. The membrane must be flat - if it wrinkles remove it and try again with a new one. It is important that all parts are clean and dry. Neither membrane nor small O-ring can be used more than once, because the membrane beds in with the cathode and is stretched a little. It will not fit perfectly a second time.

1) Unscrew ring, discard used membrane and O-ring

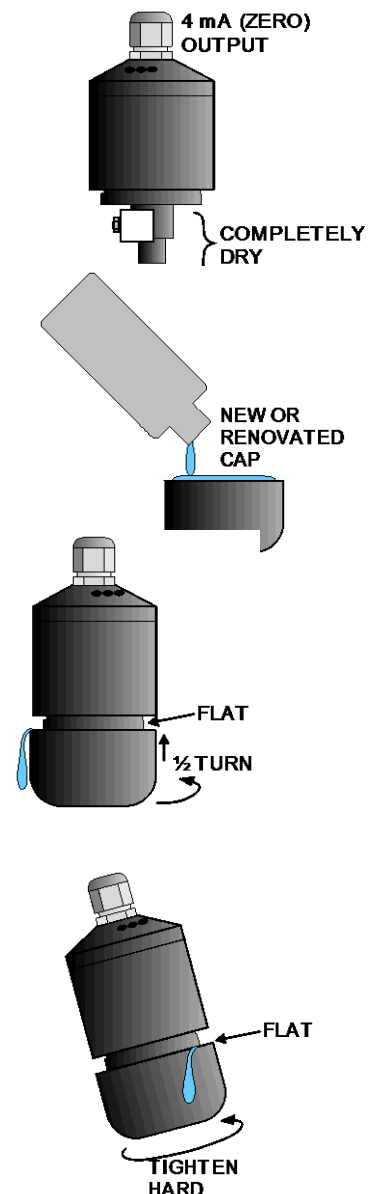
2) Clean and dry cap & ring **thoroughly**



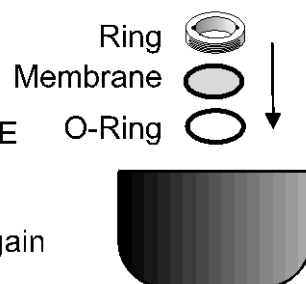
3) Assemble as shown.

THE O-RING GOES UNDER THE MEMBRANE

4) Tighten the ring. If the membrane wrinkles try again with a new membrane.



It is not recommended to fit a cap with a membrane that has been used before.



4.2 Spare Parts

Spare membranes, o-rings and electrolyte for the first few years' use are shipped with the probe, after which you can purchase more. A stock of these parts will enable you to replace a damaged membrane in a few minutes. If desired you can stock a spare probe – you will then be able to replace a probe that is accidentally destroyed, damaged or lost. A spare probe can be kept ready-to-use for years – it should be stored in a cool place. Shake it close to your ear before use – if it needs re-filling with electrolyte it is recommended that you renovate it as described above in section 4.1.

A stock of one or more spare caps will make it easy to renovate probes with damaged or “old” membranes – you can fit new membranes to the “old” caps indoors in the dry ready for next time.

You can also keep one or more spare anodes – type 3 anodes must be washed in soapy water to remove all protective oil before use.

Spare Part Numbers

Sets of 10 membranes with small O-rings:	D10PP - Membrane protector.
D10MM - For ppm (mg/l) measurements.	D10TOOL - Membrane ring removal tool.
D10MSV - For % sat & % volume measurements.	
	D10E3500 - 500 ml type 3 electrolyte.
D10AN3 - Anode for type 3 probes.	D10E31L - 1 l type 3 electrolyte.

5. SPECIFICATIONS

Dimensions and Weight:	Diameter = 58 mm, length = 88 mm. Cable length = 7 m (standard), Weight 700 g incl. cable.
Principle of Measurement:	Membrane covered galvanic cell, self polarising, self temperature-compensating.
Measuring Ranges:	DO from 0-10 mg/l to 0-40 mg/l (0-100% to 0-400% sat.). Gaseous oxygen 0-25% oxygen. Temperature (if included) -5 to +45°C
Operating Conditions:	0 to 40°C.
Minimum Flow Requirement:	Typically 1 cm/sec (for DO value 7 mg/l at 13°C).
Accuracy:	Zero drift less than +/- 0.1 mA. Measuring error less than +/- 0.1 mA when measuring temperature and pressure are the same as when calibrated. Response time: 90% of end value within 1 minute.
Loop Power Supply:	From 12 VDC at 50 ohm total loop resistance to max. 32 VDC (1050 ohm). See figure on page 2.
Warm-up time	Approx. 1 second. Depends on actual conditions.
Galvanic Separation:	1000 volts RMS input/output
Connections:	Brown lead plus, blue lead minus.
Model with temperature sensor and transmitter:	
Additional range:	4-20 mA corresponds to -5 to +45°C.
Connections temperature:	Yellow lead plus, black lead minus.
Connections oxygen:	Brown lead plus, blue lead minus.
Otherwise as above.	

Data subject to change without notice.